

Product Vision Document

SentinelCut — AI-Powered Audio/Video Profanity Censorship Platform

Document Version: 1.0 **Prepared For:** Investors, Engineering Team, Academic Reviewers **Document Owner:** Product Management **Status:** Draft for Review

1. Executive Summary

SentinelCut is an AI-driven SaaS platform that automatically detects and censors profanity, abusive language, and offensive content in audio and video files across English, Hindi, Hinglish, and mixed-language speech. It replaces manual, time-consuming audio bleeping — traditionally done frame-by-frame in tools like Adobe Premiere or Audacity — with an automated, editable, multilingual pipeline that produces broadcast-ready output in minutes instead of hours.

This document defines the product's purpose, target users, market opportunity, scope, and success criteria. It is the foundational reference that all subsequent documents (PRD, SRS, Architecture, Roadmap) will build upon.

Note on naming: "AI Profanity Video & Audio Censorship Platform" is descriptive but not brandable. We recommend **SentinelCut** (alternates considered: *CleanCast*, *MuteAI*, *CensorAI*, *PurgeTrack*) as a working product name for documentation and demo purposes. This can be finalized later without affecting technical scope.

2. Problem Statement

2.1 The Core Problem

Content creators, broadcasters, podcast editors, educational platforms, and enterprises that publish audio/video content routinely need to remove profanity or offensive language to meet:

- Platform community guidelines (YouTube, Instagram, OTT platforms)
- Broadcast/censorship regulations (India's CBFC-style norms, FCC in the US)
- Workplace/educational content standards
- Regional language sensitivities (a word innocuous in one dialect may be offensive in another)

2.2 Why Existing Solutions Fall Short

Current Approach	Limitation
Manual editing in Premiere/Audacity	Extremely slow (hours per file), requires skilled editors, doesn't scale
Existing Western censorship tools (e.g., basic beep-editors)	English-only, dictionary-based, no support for Hindi/Hinglish/code-mixed speech
Generic transcription tools	Detect speech but have no profanity-awareness, slang detection, or censorship-specific workflow
Manual review by moderation teams	High cost, inconsistent judgment, not scalable, slow turnaround

2.3 The Gap We Address

There is **no accessible, affordable, self-serve platform** that combines:

1. Multilingual speech recognition (English + Hindi + Hinglish + code-mixed)
2. Context-aware profanity detection (not just dictionary matching — handles slang, spelling variations like "fuuuuck," abbreviations like "bc," "mc," "bkl," and leetspeak like "m@darchod")
3. An editable timeline UI for human-in-the-loop correction
4. Automated, synchronized audio/video re-generation

This gap is especially pronounced for **Indian and South Asian content creators**, who make up one of the largest and fastest-growing content markets globally but are underserved by English-first tools.

3. Target Users

3.1 Primary Personas

1. Independent Content Creator ("Rohan, 24, YouTuber/Podcaster")

- Publishes vlogs/podcasts in Hinglish, needs quick "clean version" for sponsors/YouTube monetization policies.
- Pain: manually bleeping takes 2-3 hours per episode.

2. OTT/Broadcast Post-Production House ("Priya, 32, Video Editor at a media studio")

- Needs to produce a "TV-safe" cut of web-first content for broadcast.
- Pain: compliance deadlines, high editor labor cost, inconsistency across editors.

3. EdTech / Corporate L&D Team ("Ankit, 29, Learning Content Manager")

- Publishes recorded lectures/webinars with informal speaker language that needs cleanup before internal/external distribution.

4. Podcast Network / Aggregator

- Needs to produce "clean" versions of episodes at scale for platforms (e.g., Spotify) requiring explicit-content tagging accuracy or clean alternates.

3.2 Secondary Personas

- Parents/educators wanting to sanitize downloaded content for children.
 - Independent film/short-film makers preparing censor-board submission cuts.
 - Enterprises with internal recorded-meeting archives requiring redaction of inappropriate language before wider distribution.
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4. Goals

4.1 Business Goals

- Establish a working, demonstrable MVP suitable for academic evaluation and portfolio/investor presentation within one semester.
- Position the platform as a foundation for a real SaaS product post-graduation (freemium credit-based model).
- Build a differentiated moat around **Hindi/Hinglish profanity detection**, an underserved NLP niche.

4.2 User Goals

- Upload a file and receive a censored, synchronized output with minimal manual effort.
- Retain full control via a review/edit step before final export (avoid "black box" AI mistakes).
- Support both fully automatic and fully custom (user-defined word list) workflows.

4.3 Technical Goals

- Achieve high transcription and detection accuracy on code-mixed Hindi-English speech, a domain where most commercial ASR tools underperform.
 - Maintain frame-accurate audio/video sync after censorship edits.
 - Design a modular pipeline (ASR → Language ID → Profanity Detection → Audio Edit → Mux) so components (e.g., ASR engine) can be swapped as better models emerge.
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5. Vision Statement

"To make every voice broadcast-ready — instantly, accurately, and in the language people actually speak."

SentinelCut envisions a future where content creators across every language, especially underserved multilingual markets like India, can publish clean, compliant content without sacrificing speed, cost, or creative control. Just as spell-check made grammar accessible to everyone, SentinelCut aims to make **content compliance** a one-click operation.

6. Value Proposition

6.1 For Content Creators

"Upload once, get a broadcast-safe version in minutes — not hours — with a transcript-based editor that puts you in control of every edit."

6.2 Key Differentiators

Differentiator	Why It Matters
Multilingual-first design (English + Hindi + Hinglish + code-mixed)	Most competitors are English-only; this is our core wedge into an underserved market
Editable timeline, not a black box	Builds trust — users can review/override every AI decision before export
Hybrid detection (dictionary + context-aware model)	Catches slang, misspellings, leetspeak, and repeated-letter variants (e.g., "fuuuuck") that pure dictionary tools miss
Multiple censorship modes (beep, mute, custom sound, future AI voice replacement)	Flexibility for different content types (kids' content vs. broadcast vs. casual)
Self-serve SaaS workflow	No need to hire an editor or use complex NLE software

7. Competitive Analysis

Product	Category	Strengths	Weaknesses (vs. SentinelCut)
Adobe Podcast / Premiere manual workflow	Professional NLE	High precision, full creative control	Fully manual, no automated detection, steep learning curve
Descript	AI transcript-based editing	Great transcript-driven editing UX, filler-word removal	No profanity-specific detection, weak Hindi/Hinglish support, not censorship-focused
Auphonic	Audio post-processing	Good audio cleanup automation	No language/profanity detection at all
YouTube auto-captions + manual bleep	Free, built-in	Free	No censorship automation, English-biased captions, poor Hindi accuracy
Generic "profanity filter" text APIs (e.g., text-only moderation APIs)	Text moderation	Fast, cheap for text	Not built for audio/video, no timestamp/audio editing capability
Regional/manual dubbing studios	Full-service	High quality	Expensive, slow, not self-serve, not scalable for small creators

7.1 Competitive Positioning Summary

No direct competitor combines **automated multilingual ASR + profanity-specific NLP + human-editable timeline + synchronized re-render** in a self-serve product. SentinelCut's primary competitive moat is **Hindi/Hinglish/code-mixed profanity detection**, a capability that is largely absent from Western-built tools.

8. Assumptions

- Users have legal rights to the audio/video content they upload.
- Internet connectivity is available for upload/processing (cloud-based MVP; offline/on-prem is a future consideration).
- A "good enough" MVP accuracy target (e.g., 85–90% detection recall) is acceptable for v1; 100% precision is not required at launch, since the timeline editor allows

human correction.

- Free-tier cloud/ML resources (as available to students, e.g., free GPU credits, open-source models) are sufficient to build and demo the MVP.
- Users are willing to review AI-flagged content before final export at this stage of product maturity (human-in-the-loop is a feature, not a limitation).

9. Constraints

Type	Constraint
Academic/Time	Must be deliverable within a college project timeline (single/double semester)
Budget	Must primarily use free-tier/open-source tools (open-source ASR models, free-tier cloud hosting, free-tier storage)
Compute	Limited or no dedicated GPU infrastructure; must design for CPU-feasible inference where possible, with GPU as an optimization path
Team Size	Small team (student-scale), requiring a modular, incrementally buildable architecture
Data	No large proprietary Hindi/Hinglish profanity-labeled dataset exists off-the-shelf; must rely on open datasets, custom dictionaries, and augmentation
Legal	Must avoid becoming a tool for content misuse (e.g., must not enable non-consensual voice cloning); future AI voice-replacement features must include safeguards

10. Scope

10.1 In Scope (MVP and Near-Term)

- Audio/video upload (common formats: MP4, MOV, MP3, WAV, etc.)
- Automatic speech recognition with language detection (English/Hindi/Hinglish)
- Automatic + custom + combined profanity detection modes
- Beep/mute/custom-sound censorship
- Editable transcript timeline (add/remove/adjust detections)
- Synchronized final render with download (video/audio/subtitle/transcript/JSON)
- Basic authentication, dashboard, history, and (for demo purposes) a basic admin panel

10.2 Out of Scope (MVP)

- Real-time/live-stream censorship (Zoom, OBS, live broadcast)
- AI voice cloning for replacement words
- Mobile native apps
- Browser/Chrome extension, Discord/Zoom integrations
- Multi-tenant enterprise billing/roles beyond basic RBAC

(These are documented in the Future Scope section as the long-term product roadmap.)

11. Success Metrics

11.1 Product/Academic Success Metrics

- **Detection Recall:** $\geq 85\%$ of profane/offensive words correctly flagged in mixed Hindi-English test audio.
- **Detection Precision:** $\geq 80\%$ (minimize false positives that annoy users).
- **Sync Accuracy:** Final output audio/video sync drift $< 100\text{ms}$ after censorship edits.
- **Processing Time:** $\leq 2\times$ real-time duration for a standard-definition file (e.g., a 10-minute video processed in ≤ 20 minutes) on available infrastructure.
- **Task Completion Rate:** $\geq 90\%$ of test users able to complete upload $\rightarrow$ review $\rightarrow$ export flow without external help.

11.2 Business/Demo Success Metrics (if pursued beyond academics)

- Number of files processed during demo/beta period.
 - User retention across repeat uploads (proxy for usefulness).
 - Qualitative feedback score from evaluators/faculty/investors on UX and technical depth.
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12. Risks

Risk	Impact	Likelihood	Mitigation
Hindi/Hinglish ASR accuracy insufficient	High	Medium	Use best-in-class multilingual ASR (evaluated in Tech Stack doc); allow manual transcript correction in timeline editor
Profanity detection misses slang/spelling variants	Medium	High	Hybrid approach: fuzzy-matching + normalization (handle repeated letters, leetspeak) + context model
Audio/video sync breaks after edits	High	Medium	Use frame-accurate FFmpeg-based processing; extensive sync testing
Compute cost/GPU unavailability	Medium	High	Design pipeline to run on CPU-feasible open-source models for MVP; treat GPU as a performance upgrade, not a hard dependency
False positives frustrate users	Medium	Medium	Human-editable timeline before final export mitigates this by design
Scope creep (very large feature list in original brief)	High	High	Strict MVP-first phased roadmap (documented in Development Roadmap)
Misuse potential (future voice cloning)	High	Low (future feature only)	Defer AI voice cloning to future phase with explicit consent/safeguard design

13. Future Vision (3-5 Year Horizon)

SentinelCut's long-term vision extends beyond file-based censorship into a **real-time, multi-platform content compliance layer**:

- **Real-time censorship** for live streaming, video calls (Zoom/OBS integration), and podcasts recorded live.
- **Platform integrations**: YouTube, Discord, and browser extensions for in-the-moment content moderation.
- **Expanded language coverage**: regional Indian languages (Tamil, Telugu, Bengali, Marathi, Punjabi) and other code-mixed language pairs globally.
- **AI voice replacement**: natural-sounding word substitution (not just beeping) using consent-safe, non-cloning voice synthesis.

- **Enterprise content moderation dashboard:** for media companies and platforms needing bulk compliance workflows, analytics, and audit trails.
- **Developer API:** allowing third-party platforms to embed censorship-as-a-service into their own upload pipelines.
- **On-device/offline processing:** for privacy-sensitive or low-connectivity use cases.

The long-term ambition is for SentinelCut to become the **default compliance layer for multilingual audio/video content**, in the same way spell-checkers and grammar tools became default layers for text.

14. Document Approval & Next Steps

This Product Vision document forms the foundation for:

1. **Product Requirements Document (PRD)** — detailed feature requirements
2. **Software Requirements Specification (SRS)** — functional/non-functional requirements
3. **Tech Stack Justification** — detailed comparison and selection of ASR, NLP, backend, frontend, and infrastructure

Next Document: Product Requirements Document (PRD)

End of Product Vision Document